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Practical 10

a) Internetworking with routers in Cisco Packet Tracer Simulator

Design and configure a simple internetwork using a router.

In this network, a router and 2 PCs are used. Computers are connected with routers using a copper straight-through cable. After forming the network, to check network connectivity a simple PDU is transferred from PC0 to PC1

Procedure:

Step 1 (Configuring Router1):

1. Select the router and Open CLI
2. Press ENTER to start configuring Router1.
3. Type enable to activate the priveleged mode.

Router1 Command Line Interface

Router> enable
Router# config t
Enter configuration commands, one per line. End with CNTL/Z. Router (config)# interface FastEthernet 0/0
Router(configuring Router1):

1. Select the router and Open CLI
2. Press ENTER to start configuring Router1
3. Type enable to activate the priveleged mode.

Router1 Command Line Interface:

Router> enable
Router# config t
Enter configuration commands, one per line. End with CNTL/Z. Router (config)# interface FastEthernet 0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#

%LINK-5-CHANGED: Interface Gigabit Ethernet0/0 changed state to up

Router(config-if)#interface FastEthernet 0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Step 2 (Configuring PCs)

1. Assign IP Addresses to every PC in the network
2. Select the PC, Go to the desktop and select IP Configuration and assign an IP address, Default Gateway, Subnet Mask.
3. Assign the default gateway of PC0 as 192.168.10.1
4. Assign the default gateway of PC1 as 192.168.20.1

Step3 (Connecting PCs with Router):

1. Connect FastEthernet0 port of PC0 with FastEthernet0/0 port of Router1 using a copper straight-through cable
2. Connect FastEthernet0 port of PC1 with FastEthernet0/1 port of Router1 using a copper straight-through cable

Router Configuration Table:

Device Name	IP Address FastEthernet0/0	Subnet Mask	IP Address FastEthernet0/1	Subnet Mask
Router1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0

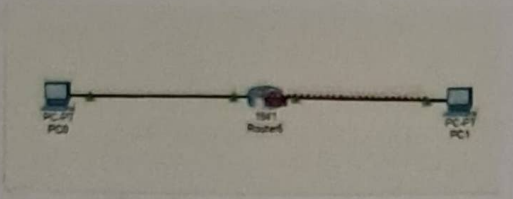
PC Configuration Table:

Device Name	IP address	Subnet Mask	Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1

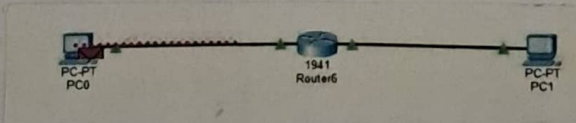
Designed Network Topology:

Simulation of Designed Network Topology:

Sending a PDU from PC0 to PC1



Acknowledgement from PC1 to PC0:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete	
	Successful	PC0	PC1	ICMP		0.000	N	0	(ed...		(delete)

Result:
Thus, interrnetworking with routers in Cisco Packet Tracer Simulator is successfully implemented.

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b) Design and configure an internetwork using wireless router, DHCP server and internet cloud.

Aim:
To design and configure an internetwork using wireless router, DHCP server and internet cloud.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC	Ethernet0	DHCP	—	192.168.0.1
Wireless Router	LAN	192.168.0.1	255.255.255.0	—
Cisco.com Wireless Router Server	Internet	DHCP	—	—
Cisco.com Server	Ethernet0	208.67.220.220	255.255.255.0	—
Laptop	Wireless0	DHCP	—	—

Part 1: Build a Simple Network in the Logical Topology Workspace

Step 1: Launch Packet Tracer

Step 2: Build the Topology

a. Add network devices to the workspace

The following devices were added:

* PC
* Laptop
* Wireless Router
* Cable Modem
* Internet Cloud
* Cisco.com Server

b. Change display names of the network devices
The display names of the network devices were configured as required.
c. Add the physical cabling between devices on the workspace
PC FastEthernet0 → Wireless Router Ethernet 1: Copper Straight-Through
Wireless Router Internet → Cable Modem Port 1: Copper Straight-Through
Cable Modem Port 0 → Internet Cloud Coaxial: Coaxial Cable
Internet Cloud Ethernet → Cisco.com server FastEthernet0: Copper Straight-Through.

Part 2: Configure the Network Devices
Step 1: Configure the Wireless Router
a. Create the wireless network on the wireless router.
Network Name (SSID): Home Network
DHCP Server: Enabled
DNS Server: 208.67.220.220
Wireless router settings were saved.

Step 2: Configure the Laptop
a. Create the wireless network on the wireless router
Network Name (SSID): Home Network
DHCP Server: Enabled
DNS Server: 208.67.220.220
Wireless router settings were saved.

a. Configure the Laptop to access the wireless network.
Wireless Module : WPC300N
Laptop was connected to the Home Network wireless network.

Step3: Configure the PC
a. Configure the PC for the wired network.
IP configuration was set to DHCP
The PC obtained an IP address in the 192.168.0.x range.
The following command was used to verify the configuration
ipconfig /all

Step4 : Configure the Internet Cloud
a. Install network modules if necessary
PT-CLOUD-NM-1CX and PT-CLOUD-NM-1CFE modules were installed if required.
b. Identify the From and To Ports
From Port : Coaxial
To Port : Ethernet
c. Identify the type of provider
Provider Network : Cable

Step 5 : Configure the Cisco.com Server
a. Configure the Cisco.com server as a DHCP server

DHCP: On
Pool Name : DHCPpool
Default Gateway : 208.67.220.220
DNS Server: 208.67.220.220
Starting IP Address : 208.67.220.1

Subnet Mask: 255.255.255.0
Maximum Number of Users: 50

b. Configure the Cisco.com server as a DNS server

DNS: On
Name: Cisco.com
Type: A Record
Address: 208.67.220.220

c. Configure the Cisco.com server Global settings.

IP Configuration: Static
Gateway: 208.67.220.1
DNS Server: 208.67.220.220

d. Configure the Cisco.com server FastEthernet0 Interface settings

IP Configuration: Static
IP Address: 208.67.220.220
Subnet Mask: 255.255.255.0

Part 3: Verify Connectivity

Step 1: Refresh the IPv4 Settings on the PC

a. Verify that the PC is receiving IPv4 configuration information from DHCP

The following commands were used to refresh the IPconfiguration
ipconfig/release
ipconfig/renew
The PC received an IP address in the 192.168.0.x range
The Subnet Mask, Default Gateway and DNS Server were also obtained through DHCP.

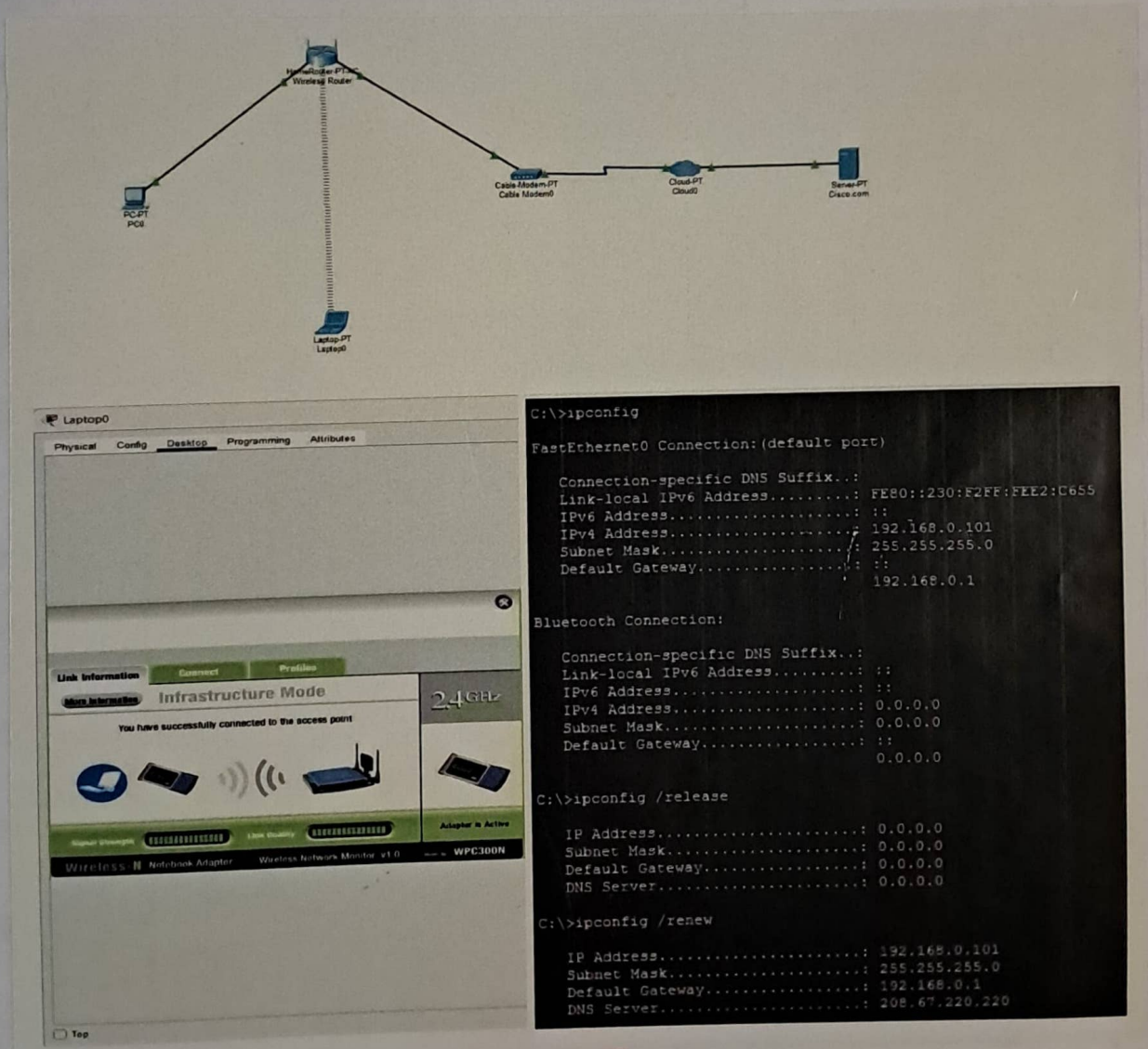
b. Test Connectivity to the Cisco.com server from the PC

The following command was used:

ping Cisco.com

Four successful replies were received from 208.67.220.220

Connectivity between the PC and Cisco.com server was successfully verified.





```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address.................: 0030.F2E2.C655
   Link-local IPv6 Address.........: FE80::230:F2FF:FEE2:C655
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.0.101
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.0.1
   DHCP Servers....................: 192.168.0.1
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-1E-2C-08-A7-00-30-F2-E2-C6-55
   DNS Servers.....................: ::
                                     208.67.220.220

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address.................: 0000.0CCD.1831
   Link-local IPv6 Address.........: ::

C:\>ping 192.169.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 208.67.220.220

Pinging 208.67.220.220 with 32 bytes of data:

Reply from 208.67.220.220: bytes=32 time=26ms TTL=127
Reply from 208.67.220.220: bytes=32 time=10ms TTL=127
Reply from 208.67.220.220: bytes=32 time=30ms TTL=127
Reply from 208.67.220.220: bytes=32 time=52ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 52ms, Average = 29ms
```

```
C:\>ping Cisco.com

Pinging 208.67.220.220 with 32 bytes of data:

Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=16ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 4ms
```

Result:

Thus, the internetwork using a wireless router, DHCP server and Internet cloud was successfully configured. Connectivity between the PC and Cisco.com server was successfully verified using the Ping command.